

PROGRAM:

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#include<stdio.h>

#include<stdlib.h>

struct BST {
    int data;

    struct BST *lchild;

    struct BST *rchild;
};

typedef struct BST * NODE; NODE create() {
    NODE temp;

    temp = (NODE) malloc(sizeof(struct BST));

    printf("\nEnter The value: ");

    scanf("%d", &temp->data);

    temp->lchild = NULL;

    temp->rchild = NULL;

    return temp;
}

void insert(NODE root, NODE newnode);

void inorder(NODE root);

void preorder(NODE root);

void postorder(NODE root);

void search(NODE root);

void insert(NODE root, NODE newnode) {
    /*Note: if newnode->data == root->data it will be skipped.    No duplicate nodes are allowed */

    if (newnode->data < root->data)    {
        if (root->lchild == NULL)
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        root->lchild = newnode;

    else

        insert(root->lchild, newnode);
}

if (newnode->data > root->data) {

    if (root->rchild == NULL)

        root->rchild = newnode;

    else

        insert(root->rchild, newnode);

}

}

void search(NODE root) {

    int key;

    NODE cur;

    if(root == NULL) {

        printf("\nBST is empty.");

        return;

    }

    printf("\nEnter Element to be searched: ");

    scanf("%d", &key);

    cur = root;

    while (cur != NULL) {

        if (cur->data == key) {

            printf("\nKey element is present in BST");

            return;

        }

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        if (key < cur->data)
            cur = cur->lchild;
        else
            cur = cur->rchild;
    }

    printf("\nKey element is not found in the BST");
}

void inorder(NODE root) {
    if(root != NULL)    {
        inorder(root->lchild);
        printf("%d ", root->data);
        inorder(root->rchild);
    }
}

void preorder(NODE root) {
    if (root != NULL)    {
        printf("%d ", root->data);
        preorder(root->lchild);
        preorder(root->rchild);
    } }

void postorder(NODE root) {
    if (root != NULL)    {
        postorder(root->lchild);
        postorder(root->rchild);
        printf("%d ", root->data);
    } }

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void main() {

    int ch, key, val, i, n;

    NODE root = NULL, newnode;

    while(1)    {

        printf("\n~~~~~BST MENU~~~~~");

        printf("\n1. Create a BST");

        printf("\n2. Display");

        printf("\n3. Search");

        printf("\n4. Exit");

        printf("\nEnter your choice: ");

        scanf("%d", &ch);

        switch(ch)    {

            case 1:

                printf("\nEnter the number of elements: ");

                scanf("%d", &n);

                for(i=1;i<=n;i++)    {

                    newnode = create();

                    if (root == NULL)

                        root = newnode;

                    else

                        insert(root, newnode);

                }

                break;

            case 2:

                if (root == NULL)

                    printf("\nTree Is Not Created");

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        else
        {
            printf("\nThe Preorder display : ");
            preorder(root);
            printf("\nThe Inorder display : ");
            inorder(root);
            printf("\nThe Postorder display : ");
            postorder(root);
        }
        break;
    case 3:
        search(root);
        break;
    case 4:
        exit(0);
    }
}
}

```

OUTPUT:

~~~~~BST MENU~~~~~

1. Create a BST
2. Display
3. Search
4. Exit

Enter your choice: 1

Enter the number of elements: 7

Enter The value: 12

Enter The value: 35

Enter The value: 13

Enter The value: 24

Enter The value: 76

Enter The value: 24

Enter The value: 64

~~~~BST MENU~~~~

1. Create a BST

2. Display

3. Search

4. Exit

Enter your choice: 2

The Preorder display : 12 35 13 24 76 67 64

The Inorder display : 12 13 24 35 64 67 76

The Postorder display : 24 13 64 67 76 35 12

~~~~BST MENU~~~~

1. Create a BST

2. Display

3. Search

4. Exit

Enter your choice: 3

Enter Element to be searched: 64

Key element is present in BST

~~~~BST MENU~~~~

1. Create a BST

2. Display

3. Search

4. Exit

Enter your choice: 3

Enter Element to be searched: 1

Key element is not found in the BST

~~~~BST MENU~~~~

1. Create a BST

2. Display

3. Search

4. Exit

Enter your choice: 4